

Bridge Day 14 INSTRUCTOR KEY

List Indexing, Mutation, and Final-State Summaries

Learning sequence: Predict - Trace - Explain - Test - Interpret

Thursday, July 23, 2026 • 20 points • target 19 minutes

Name		UIN		Section	
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Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: for/range, len, indexes, list mutation, final-state summaries

Scaffold level: complete indexed mutation and summary program

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.25, 18.26, 18.27.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Trace mutation and index-dependent updates. - 4 points

```
values = [4, -2, 7, -1]
changed = 0

for index in range(len(values)):
    if values[index] < 0:
        values[index] = 0
        changed = changed + 1
    values[index] = values[index] + index

print(values, changed)
```

Question: Give the list and changed state after every index, then write exact output.

Model response: Index states are 0 -> [4, -2, 7, -1], 0; 1 -> [4, 1, 7, -1], 1; 2 -> [4, 1, 9, -1], 1; 3 -> [4, 1, 9, 3], 2. Exact output: [4, 1, 9, 3] 2.

Item 2. Repair a loop that changes only its temporary value. - 4 points

```
values = [-2, 4, -1]
total = 0

for value in values:
    if value < 0:
        value = 0
    total = total + value

print(values, total)
```

Question: Give actual output, explain why the list does not change, and rewrite the loop header/body to mutate positions.

Model response: The local name value becomes 0 for negative passes, so total is 4, but no list position is assigned. Actual output: [-2, 4, -1] 4. Repair by looping over range(len(values)) and assigning values[index] = 0 when negative.

Complete program - 12 points

- Read threshold as an integer and use readings = [12, -3, 7, 20, 7].
- Loop by index and replace every negative value with 0.
- Using final values, compute total and the count at or above threshold.
- Find the index of the first largest final value without max or index.
- Print Final, Total, At or above, and Largest index on separate labeled lines.

Program workspace

```
Model program
threshold = int(input())
readings = [12, -3, 7, 20, 7]
total = 0
at_or_above = 0
largest = readings[0]
largest_index = 0

for index in range(len(readings)):
    if readings[index] < 0:
        readings[index] = 0
    value = readings[index]
    total = total + value
    if value >= threshold:
        at_or_above = at_or_above + 1
    if value > largest:
        largest = value
        largest_index = index

print(f"Final: {readings}")
print(f"Total: {total}")
print(f"At or above: {at_or_above}")
print(f"Largest index: {largest_index}")
```

Test input, expected result, and why this test matters

Reasoning and test evidence Mutation occurs before all summaries, so total, threshold count, and largest state describe the final list. Strict > preserves the first largest index if a tie occurs. With threshold 10, the final list is [12, 0, 7, 20, 7], total is 46, at_or_above is 2, and largest_index is 3.

Scoring guidance: 2 input/initialization, 2 indexed mutation, 3 final total/count, 2 first-largest tracking, 1 exact output, 2 tests/explanation.